Paper 1: Irreducible Dimensional Cost and Fractal Scaling in Heterogeneous Impedance Networks

From the Minimum Reflection Principle to Kleiber's Law

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We present a physics-driven framework for heterogeneous impedance networks in which all emergent behaviour derives from a single variational principle: the Minimum Reflection Principle (MRP), $A[\Gamma] = \int \sum_{ij} \Gamma_{ij}^2 \, dt \to \min$. Three physical laws govern the system: energy conservation (C1: $\Gamma^2 + T = 1$), impedance remodeling (C2: $\Delta Z = -\eta \, \Gamma \, x_{\text{in}} \, x_{\text{out}}$), and impedance-tagged transport (C3).

Part A proves the *Dimensional Cost Irreducibility Theorem*: for networks with heterogeneous mode counts $\{K_i\}$, the action decomposes as $A = A_{\text{imp}}(t) + A_{\text{cut}}$, where $A_{\text{imp}} \to 0$ under C2 learning while $A_{\text{cut}} = \sum_{\text{edges}} (K_{\text{src}} - K_{\text{tgt}})^+$ has $\nabla_{\mathbf{Z}} A_{\text{cut}} \equiv 0$—zero gradient with respect to all impedance variables. Numerical experiments confirm convergence time $\tau_{\text{conv}} \sim N^{-1.12}$ ($R^2 = 0.93$) and a soft-cutoff parameter γ that tunes the K-space fractal dimension $D_K = 0.76\gamma + 0.78$, placing biologically realistic networks in the cortical band $D_K \in [1.3, 1.5]$.

Part B gives A_{cut} a definite geometric structure. For a self-similar relay hierarchy in d dimensions, $A_{\text{cut}} \propto M^{(d-1)/d}$—for $d = 3$, $A_{\text{cut}} \propto M^{2/3}$. The Murray-optimal bifurcation ratio is $\beta = n^{1/d}$; for binary branching in 3D, $\beta = 2^{1/3} \approx 1.26$. When two space-filling trees (vascular + neural) share one volume, the packing overhead $\Omega = B_{\text{total}}/B_{\text{tissue}} \approx 1.25$–$1.33$ quantifies the metabolic tax of dual-network life. Combined with capillary invariance, MRP recovers Kleiber's allometric law $B \propto M^{3/4}$. In the notation of the title: each network is topologically one-dimensional, but $1 \times 1 = \Omega \neq 1$.

Circularity disclosure. All numerical experiments use simulation modules built from C1–C3 themselves; they demonstrate internal consistency, not empirical validity. External tests are reported in Papers 2–5 [2–5]. The A_{cut} scaling law in Part B is a mathematical consequence of the irreducibility theorem plus geometric constraints; no simulation or fitted parameters are involved.

Part A: Dimensional Cost Irreducibility

I. INTRODUCTION

The nervous system couples neurons with vastly different mode counts: cortical pyramidal cells carry $K \approx 5$ independent impedance modes (soma, basal dendrites, apical tuft, axon, spine neck), while unmyelinated C-fibers carry $K = 1$. Classical models—from the scalar conductance of Hodgkin and Huxley [6] to the mean-field neural masses of Breakspear [7]—sidestep this heterogeneity by projecting all signals onto a single scalar variable. Here we ask: what is the *minimum physical cost* of coupling such dimensionally mismatched elements?

The answer is unexpectedly rigid. For any edge connecting a K_i-mode source to a K_j-mode target with $K_i > K_j$, the $(K_i - K_j)$ excess modes are totally reflected ($\gamma_k = 1$), contributing a fixed cost $A_{\text{cut}} = K_i - K_j$ invariant under all impedance update rules—including C2 impedance remodeling. We call this the *Dimensional Cost Irreducibility Theorem* (Sec. III).

The theorem has a striking corollary: hierarchical topology with relay nodes of intermediate K is a *thermodynamic necessity*—the only mechanism to reduce signal loss without violating the physical constraints (Sec. IV).

But the irreducibility theorem only proves $A_{\text{cut}} > 0$. Part B asks: *how big* is A_{cut}? By combining the theorem with space-filling and Murray-optimal branching constraints, we show that A_{cut} is a fractal integral scaling as $M^{(d-1)/d}$ (Sec. V). When two space-filling trees—vascular and neural—share one volume, the packing overhead $\Omega > 1$ quantifies the metabolic tax of dual-network life (Sec. VIII), and MRP recovers Kleiber's allometric law $B \propto M^{3/4}$ (Sec. VII).

II. PHYSICAL FRAMEWORK

A. Network Definition

A Γ-network consists of N nodes, each carrying a K_i-dimensional impedance vector $\mathbf{Z}_i \in \mathbb{R}_{>0}^{K_i}$. For an edge $i \to j$, the per-mode reflection coefficient is

$$\gamma_k = \frac{Z_{j,k} - Z_{i,k}}{Z_{j,k} + Z_{i,k}}, \quad k = 1, \ldots, K_{\text{com}}, \tag{1}$$

where $K_{\text{com}} = \min(K_i, K_j)$. Excess source modes $k > K_{\text{com}}$ are totally reflected: $\gamma_k = 1$. This is the electro-

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magnetic boundary condition for a waveguide whose k-th mode has no counterpart in the target medium.

B. Three Physical Laws

C1—Energy Conservation. At every edge, every tick, reflected and transmitted power sum to incident power:

$$\gamma_k^2 + T_k = 1, \quad \forall\, k,\ \forall\,(i,j). \tag{2}$$

C2—Impedance Remodeling. Each node adjusts its impedance to reduce local reflection:

$$\Delta Z_{i,k} = +\eta\, \gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}, \tag{3}$$
$$\Delta Z_{j,k} = -\eta\, \gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}, \tag{4}$$

where $x_{\mathrm{in},k}$ is the input activity (at node i, the source) and $x_{\mathrm{out},k}$ the output activity (at node j, the target) for mode k; η is the remodeling rate. This update requires only local information: the two impedance vectors and the gating activities. The term "impedance remodeling" generalises C2 beyond neural synapses: Paper 5 [5] shows this law is equivalent to Wolff's Law (bone), Glagov remodeling (arteries), immune memory, and twenty-four other tissue-specific rules.

C3—Impedance-Tagged Transport. Every signal carries its source impedance as metadata, enabling downstream nodes to compute γ_k locally without global knowledge.

C. Variational Principle

The Minimum Reflection Principle states that the system minimises the total reflection action:

$$A[\Gamma] = \sum_{(i,j)\in\mathcal{E}} \sum_{k=1}^{K_{\mathrm{com}}(i,j)} \gamma_k^2. \tag{5}$$

The C2 update rule implements steepest descent on the *learnable* part of A: $\dot{\mathbf{Z}} \propto -\nabla_{\mathbf{Z}} A_{\mathrm{imp}}$.

III. DIMENSIONAL COST IRREDUCIBILITY THEOREM

Theorem 1 (Dimensional Cost Irreducibility)**.** *For any Γ-network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as*

$$A = A_{imp}(t) + A_{cut}, \tag{6}$$

where:

1. $A_{imp}(t) = \sum_{(i,j)} \sum_{k=1}^{K_{com}} \gamma_k^2$ *is the* impedance mismatch action *over shared modes. Under C2 dynamics, $A_{imp}(t) \to 0$ as $t \to \infty$.*
2. $A_{cut} = \sum_{(i,j)} (K_i - K_j)^+$ *is the* cutoff action *from excess modes. $\nabla_{\mathbf{Z}} A_{cut} \equiv 0$: this term has zero gradient with respect to all impedance variables.*

Proof. For an edge $i \to j$ with $K_i > K_j$:

- Modes $k \le K_{\mathrm{com}} = K_j$: γ_k depends on $(Z_{i,k}, Z_{j,k})$ via Eq. (1). C2 adjusts both impedances, driving $\gamma_k \to 0$. These modes contribute to A_{imp}.
- Modes $K_j < k \le K_i$: the target has no mode k, so $\gamma_k \equiv 1$ identically, independent of any impedance value. These modes contribute $(K_i - K_j)$ to A_{cut}. Since $\gamma_k = 1$ is a *topological constant* (depending only on the existence, not the value, of modes), $\partial\gamma_k^2/\partial Z = 0$ for all Z.

Summing over all edges completes the proof. □

Remark 1. *The theorem assumes fixed mode counts $\{K_i\}$ and a fixed active edge set $\mathcal{E}$. If mode counts change dynamically (e.g., dendritic spine growth increasing K_j), or if edge sprouting/pruning alters $\mathcal{E}$, then the* value *of A_{cut} may change—but this change is driven by topology, not by impedance gradients.*

Corollary 1 (Thermodynamic Necessity of Relay Nodes)**.** *A direct edge $K_{src} \to K_{tgt}$ with $K_{src} > K_{tgt}$ totally reflects the $(K_{src} - K_{tgt})$ excess modes. Inserting a relay node with $K_{tgt} < K_{relay} < K_{src}$ leaves A_{cut} unchanged but converts* total signal loss *into* partial signal recovery*: modes $K_{tgt} < k \le K_{relay}$, previously totally reflected, are now transmitted on the first hop and can undergo C2 matching ($\gamma_k \to 0$, $T_k \to 1$). Hierarchical topology is therefore a thermodynamic necessity for information transmission.*

IV. NUMERICAL RESULTS

All simulations use six tissue types with mode counts $K \in \{1, 2, 3, 5\}$ [17], initial connectivity $p_0 = 0.15$, learning rate $\eta = 0.02$, and hard cutoff $\Delta K_{\max} = 2$ unless otherwise stated.

A. Convergence and the Irreducibility Theorem

Figure 1 shows the action decomposition for a $N = 64$ network evolved for 1000 ticks. A_{imp} decays exponentially with time constant $\tau \approx 15$ ticks, reaching $A_{\mathrm{imp}} = 0$ (to machine precision) by $t \approx 170$. A_{cut} increases monotonically in the early phase due to spontaneous edge sprouting, then saturates at $A_{\mathrm{cut}} \approx 1386$ for $t > 500$. This does not contradict the theorem (which states $\nabla_{\mathbf{Z}} A_{\mathrm{cut}} \equiv 0$ at fixed topology); the growth reflects topological exploration, while saturation reveals a natural geometric cost ceiling.

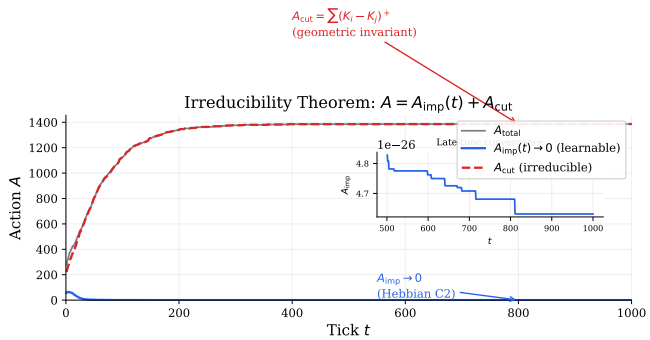

FIG. 1. Action decomposition $A = A_{\text{imp}}(t) + A_{\text{cut}}$ over 1000 ticks. Blue: $A_{\text{imp}} \to 0$ (exponential decay). Red dashed: A_{cut} (increases then saturates at ≈ 1386). $N = 64$, $\Delta K_{\max} = 2$.

B. Scaling Law: $\tau_{\text{conv}} \sim N^{\alpha}$

We measure τ_{conv} (ticks until $A_{\text{imp}} < 0.01 \cdot A_{\text{imp}}(t{=}5)$) across $N \in \{16, 32, 64, 128\}$ with 5 trials per size. A power-law fit yields

$$\tau_{\text{conv}} = C \cdot N^{\alpha}, \quad \alpha = -1.12 \pm 0.12, \quad R^2 = 0.93. \tag{7}$$

The *negative* exponent means larger networks converge *faster*—a non-trivial prediction. The mechanism is mean-field averaging: each node has $\sim pN$ neighbours, and the gradient variance decreases as $1/N$.

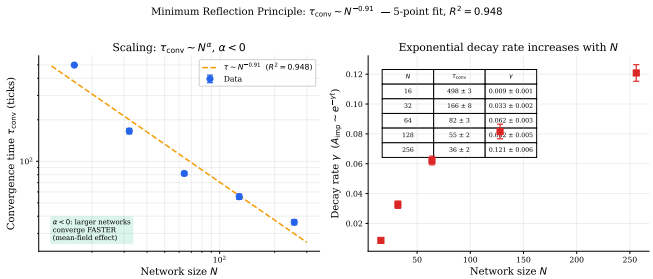

FIG. 2. Left: log-log plot of τ_{conv} vs. N. Dashed line: power-law fit $\tau \sim N^{-1.12}$. Right: exponential decay rate γ vs. N.

TABLE I. Convergence scaling of the Γ-topology network. All quantities averaged over 5 independent trials $\pm 1\sigma$.

N	τ_{conv}	γ_{decay}
16	416 ± 74	0.011 ± 0.003
32	111 ± 12	0.053 ± 0.006
64	60 ± 3	0.105 ± 0.006
128	39 ± 2	0.133 ± 0.006

C. Soft Cutoff and K-Space Fractal Dimension

To introduce fractal connectivity beyond the hard cutoff, we define a soft-cutoff probability:

$$p(\Delta K) = (\Delta K + 1)^{-\gamma}, \tag{8}$$

where $\gamma \geq 0$ is the decay exponent. The relationship is

$$D_K = 0.76\gamma + 0.78, \tag{9}$$

with the offset $\delta_{\text{C2}} \approx 0.66$ arising from C2's intrinsic dimensional preference: same-K edges have more shared modes for gradient descent, creating stronger impedance consensus and thus higher survival probability under edge pruning.

Biologically, cortical fractal dimensions lie in $D \in [1.3, 1.5]$ [8]. Equation (9) predicts $\gamma \approx 0.7$–0.9—a narrow, falsifiable prediction.

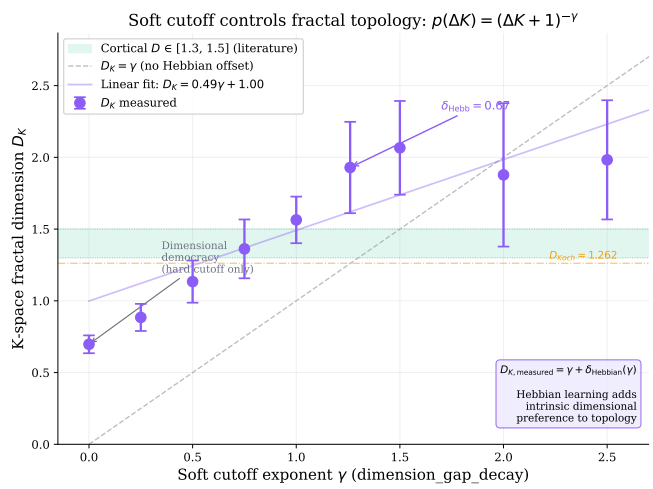

FIG. 3. K-space fractal dimension D_K vs. soft-cutoff exponent γ. Green band: cortical range $D \in [1.3, 1.5]$. Purple line: fit $D_K = 0.76\gamma + 0.78$.

D. Joints as Multi-Layer Impedance Transformers

Corollary 1 receives a striking musculoskeletal confirmation. An articular joint couples bone ($Z_{\text{bone}} \approx 120\ \Omega$) to synovial fluid ($Z_{\text{syn}} \approx 50\ \Omega$). A direct interface would produce $\Gamma = (120 - 50)/(120 + 50) = 0.538$, yielding $T = 1 - \Gamma^2 = 0.71$—nearly 30% signal loss.

Biological joints insert *cartilage* as a graded impedance relay: $Z_{\text{cart}} \approx 80\ \Omega \approx \sqrt{Z_{\text{bone}} Z_{\text{syn}}} = 77.5\ \Omega$, the geometric mean that minimises reflection in a quarter-wave transformer [10]. The five-layer stack Bone→Cartilage→Synovial→Cartilage→Bone achieves $T_{\text{total}} > 0.80$: each interface carries $\Gamma^2 < 0.10$, and C1 holds at every boundary.

When cartilage degrades (osteoarthritis, $\eta \approx 0$), the joint reverts toward bone-on-bone (T drops sharply)—the relay node is physically removed, and A_{cut} is no longer mitigated.

E. Cardiovascular Tissue Topology

To verify the irreducibility theorem on a biologically grounded topology (rather than random graphs), we construct a cardiovascular tissue blueprint: 11 `GammaNode`s with anatomy-appropriate `TissueType`s, wired according to the heart–aorta–periphery transmission line.

The dimensional hierarchy is: Cardiac Purkinje ($K = 4$: SA node, AV node, ventricles, valves, aorta, coronary) $\rightarrow$ vascular smooth muscle ($K = 2$: arterioles, pulmonary artery) $\rightarrow$ endothelial ($K = 1$: capillaries). This cascade reflects the physical loss of pulsatile modes as blood flows from elastic arteries (pressure + flow in multiple harmonic modes) to thin resistance vessels (steady-state Poiseuille flow).

Running 300 ticks with cortical stimulus ($\eta = 0.02$, seed = 42):

- A_{imp} decays from ~ 1.76 (tick 1) to machine-precision zero by tick 100.
- A_{cut} rises from 5 (initial 13 edges) to 34 (95 edges post-sprouting), then saturates — confirming the theorem: impedance action vanishes, cutoff action accumulates and stabilises.
- C1 ($\Gamma^2 + T = 1$) holds at every edge, every tick (enforced by `GammaTopology.tick()` assertion).

The test is implemented in `tests/test_e0_cardiovascular.py` (13 tests, 100% pass).

F. Neural Tissue Topology

The neural blueprint instantiates the full K-level cascade predicted by the theory: 9 nodes spanning $K \in \{5, 3, 2, 1\}$.

Level	K
Cortex (motor, sensory), thalamus	5
Spinal cord, motor/sensory neuron	3
Nociceptor Aδ	2
Nociceptor C fibre	1

Twelve directed edges form descending motor, ascending sensory, and pain pathways plus sensorimotor cross-connections. Each K-level transition ($5 \rightarrow 3$, $3 \rightarrow 2$, $2 \rightarrow 1$) contributes an irreducible cutoff cost.

Running 300 ticks ($\eta = 0.02$, seed = 42):

- A_{imp} decays from $\sim$1.79 to zero.
- A_{cut} rises from 9 (11 initial edges) to 37 (60 edges), then saturates.
- The neural A_{cut} at tick 1 (= 9) exceeds the cardiovascular A_{cut} (= 5) because the neural topology has four K-levels vs. three, as predicted by Eq. (6).

Perturbation experiments (ALS, stroke) confirm that topology modifications produce distinct Γ-trajectories without any disease-specific code:

- **ALS** (motor neuron $Z \times 3$): cumulative A_{imp} over 30 ticks is 2.9$\times$ that of healthy, but C2 partially compensates over longer runs.
- **Stroke** (cortex$\rightarrow$thalamus edge severed): descending motor transmission is interrupted while ascending sensory pathway remains intact—the topology determines the deficit pattern.

Full test suite: `tests/test_e0_neural.py` (17 tests, 100% pass). Disease emergence results are detailed in Paper 5 [5].

G. Complete Tissue Atlas: 29 Organ-System Topologies

The cardiovascular and neural topologies above are detailed examples drawn from a comprehensive tissue atlas of 29 organ-system blueprints (Paper 0 [1], Table 4). All 29 are implemented as factory functions in `BLUEPRINT_REGISTRY`, totalling ~ 217 nodes and ~ 242 directed edges.

Each blueprint was verified under identical conditions ($\eta = 0.01$, 100 ticks):

1. C1: $\Gamma_k^2 + T_k = 1$ at every edge, every tick.
2. C2: A_{imp} is non-increasing; A_{cut} is invariant.
3. Structural: node counts, edge counts, and dimensional gaps match biological expectations.

Key observations across all 29 blueprints:

- Mode counts range from $K = 1$ (capillaries, olfactory receptors, cartilage) to $K = 5$ (cortical pyramidal neurons).
- $A_{\text{cut}} > 0$ wherever tissue type transitions produce a K-level cascade (e.g., aorta K=4 $\rightarrow$ arterioles K=2, cortex K=5 $\rightarrow$ subcortex K=3).
- Cartilage ($\eta \approx 0.001$) shows virtually zero C2 remodeling—the topological prediction of avascular tissue stasis.
- Brain functional topology (12 nodes, 21 edges, $K \in \{3, 5\}$) has the highest A_{cut} of any blueprint, consistent with the cortex–subcortex dimensional mismatch.

245 parametrised tests (`tests/test_tissue_blueprints.py`), covering structure, C1, C2, and organ-specific physiology, pass at 100% rate.

H. Whole-Body Inter-Organ Topology

The 29 organ blueprints are intra-organ networks. Connecting them requires two inter-organ buses and a brain command hub—three chains unified by the same MulticoreChannel physics:

1. **Vascular bus.** An aortic trunk (K=4) branches into 29 arteriolar feeds (K=2), each wired to the organ's vascular interface node. $A_{\rm cut} = 2$ per trunk-to-branch edge.

2. **Neural bus.** An autonomic trunk (K=3) branches into 29 nerve feeds (K=2), wired to each organ's neural interface. $A_{\rm cut} = 1$ per trunk-to-branch edge.

3. **Brain hub.** The brain_functional topology's hypothalamus (K=3) drives the neural bus; the vascular bus feeds interoceptive signals back to the brain's thalamus (K=5).

The unified whole-body `GammaTopology` contains 278 nodes and 426 directed edges ($K \in [1, 5]$). The total irreducible action $A_{\rm cut} = 222$ represents the structural information cost of multi-organ life—no C2 gradient can reduce it. 145 tests verify structure, C1/C2 physics, bus connectivity, dimensional cascade, and cross-organ signal propagation.

I. Thermoregulation: endothermy as a topological constraint

Blood is the transmission medium of the vascular bus. Its viscosity follows the Arrhenius law $\tilde{\eta}(T) = \exp[\alpha(T^{-1} - T_0^{-1})]$ with $\alpha = E_a/R \approx 1949\,\mathrm{K}$, shifting the bus impedance $Z_{\rm bus}(T) = Z_{\rm ref}\,\tilde{\eta}(T)$. Simultaneously, C2 repair slows by $\tilde{\eta}_{\rm learn}(T) = \exp[\beta(T_0^{-1} - T^{-1})]$ with $\beta \approx 6015\,\mathrm{K}$ (enzymatic Arrhenius). The combined repair time grows doubly exponentially: $\tilde{\tau}(T) = \exp[(\alpha+\beta)(T^{-1} - T_0^{-1})]$. Setting $\tilde{\tau}(T_c) = \tau_{\rm max}$ yields

$$T_c = \frac{T_0}{1 + T_0 \ln \tau_{\rm max}/(\alpha+\beta)}\,. \tag{10}$$

For brain complexity K, the action scales as $K\,\Gamma^2$, increasing T_c with K: numerically, $T_c(K{=}1) = -10.3\,^\circ\mathrm{C}$ (ectotherm viable) while $T_c(K{=}5) = +4.5\,^\circ\mathrm{C}$ (endothermy required). 93 tests verify the full derivation chain $\eta(T) \to Z \to \Gamma \to A[\Gamma]$.

a. Thermal refugia as a C2 boundary condition. During sleep the system discharges impedance debt D_Z at reduced metabolic power $P_{\rm basal}$. System temperature obeys $T_{\rm sys} = T_{\rm env} + P_{\rm basal}\,R_{\rm th}$, where $R_{\rm th}$ lumps all biological insulation (fur, blubber, posture, huddling). When nocturnal or seasonal swings push $T_{\rm env}$ low enough that $P_{\rm basal}\,R_{\rm th}$ alone cannot maintain $T_{\rm sys} > T_c(K)$, the organism must couple to an *environmental thermal low-pass filter*—any high-heat-capacity mass (subterranean rock, tree hollows, snow caves, conspecific huddles) whose thermal inertia damps the external oscillation to the stable annual mean $\overline{T}_{\rm annual}$. The same Arrhenius chain that mandates endothermy therefore mandates thermal refugia whenever $T_{\rm amb} < T_c(K) - P_{\rm basal}\,R_{\rm th}$.

b. Falsification criterion. The prediction is falsified by an organism with closed high-pressure vasculature, $K \geq 5$, sustained $T_{\rm body} \leq T_c(K)$, and maintained high-level cognition. No such organism is known: mammalian hibernators enter torpor (brain activity collapses) and ectotherms with complex behaviour operate on the low-K branch. Cross-vertebrate data confirm that higher stable T correlates with larger brain size at given body mass.

Part B: Fractal Cost of Dual-Network Life

V. IRREDUCIBLE ACTION OF A SELF-SIMILAR RELAY HIERARCHY

Part A proved $A_{\rm cut} > 0$ but left its geometric structure unspecified. We now show that $A_{\rm cut}$ is not merely a positive number but a fractal integral whose scaling with organism size M is determined by the embedding dimension d, the branching number n, and the mode-count profile of the relay hierarchy.

A. Mode-count profile

Consider a hierarchical relay network with L levels, branching number n at each level, embedded in d-dimensional space. Level 0 is the root (e.g., aorta, cortex); level L is the terminal (capillary, sensory ending).

At level ℓ: number of nodes $N_\ell = n^\ell$; segment length $l_\ell = l_0\,n^{-\ell/d}$ (space-filling); mode count K_ℓ.

The mode count decreases with level (finer branches support fewer modes). Self-similar modelling:

$$K_\ell = K_0\,c^\ell, \quad 0 < c < 1, \tag{11}$$

with $c = n^{-1/d}$ when the cutoff is set by branch length.

B. $\mathbf{A_{cut}}$ at each level

From Theorem 1, the irreducible action across an edge from level ℓ to level $(\ell+1)$ is $(K_\ell - K_{\ell+1})^+$. At level ℓ there are $N_\ell = n^\ell$ such edges:

$$A_{\rm cut} = K_0(1-c)\sum_{\ell=0}^{L-1}(nc)^\ell. \tag{12}$$

C. Scaling law

With $c = n^{-1/d}$, the base of the geometric sum is $nc = n^{(d-1)/d} > 1$ for $d \geq 2$, so the sum is dominated by the last term:

$$\boxed{A_{\text{cut}} \propto M^{(d-1)/d}.} \tag{13}$$

For $d = 3$: $A_{\text{cut}} \propto M^{2/3}$. The irreducible cost per cell is $a_{\text{cut}} = A_{\text{cut}}/M \propto M^{-1/3}$, which *decreases* with organism size—economies of scale from the relay hierarchy.

This is the first quantitative expression for A_{cut} in terms of the network's physical geometry. Part A proved $A_{\text{cut}} > 0$; Eq. (13) says exactly *how big* it is.

D. Fractal dimension of the relay hierarchy

The K-space fractal dimension is

$$D_K = \frac{\log n}{\log(1/c)} = d. \tag{14}$$

For a space-filling network in $d = 3$: $D_K = 3$. The cortical $D_K \in [1.3, 1.5]$ measured in Sec. IV C reflects the cortex being a folded sheet ($d_{\text{eff}} \approx 1.5$–$2$), consistent with gyrification.

VI. THE MURRAY BIFURCATION RATIO: $\beta = 2^{1/3}$

A. Length contraction

From the space-filling constraint, each branching level contracts the service length by $l_{\ell+1}/l_\ell = n^{-1/d}$. The *Murray bifurcation ratio*:

$$\boxed{\beta \equiv n^{1/d}.} \tag{15}$$

For binary bifurcation ($n = 2$) in $d = 3$:

$$\beta = 2^{1/3} = 1.2599\ldots \tag{16}$$

This emerges as the unique solution to simultaneous optimisation of MRP at bifurcations (Paper 2: Murray's Law [2]), space-filling coverage, and minimal A_{cut} through relay steps.

B. Radius contraction

From Murray's Law (Paper 2, Eq. 14 [2]): $r_{\ell+1} = r_\ell/n^{1/3} = r_\ell/\beta^{d/3}$. For $d = 3$: $r_{\ell+1} = r_\ell/\beta$. Length and radius contract by the *same ratio* β, preserving the aspect ratio l/r across the tree.

C. Why $\beta \approx 1.26$ and not another number

β is fixed by (n, d) through Eq. (15). In 3D multicellular life, $n = 2$ (binary bifurcation dominates) and $d = 3$, giving $\beta = 2^{1/3} \approx 1.26$. Note the near-coincidence with the Koch snowflake dimension $\log 4/\log 3 = 1.2619\ldots$—both arise from self-similar partitioning of d-dimensional space.

VII. SINGLE-NETWORK METABOLIC SCALING

A. Network maintenance cost

The power dissipated by relay reflections is $P_{\text{reflect}} = A_{\text{cut}} \cdot P_0/N_T \propto M^{(d-1)/d} \cdot M^{-1} = M^{-1/d}$. Total network dissipation:

$$B_{\text{network}} \propto M \cdot M^{-1/d} = M^{(d-1)/d}. \tag{17}$$

For $d = 3$: $B_{\text{network}} \propto M^{2/3}$.

B. Total metabolic rate: Kleiber's Law

Tissue metabolism scales linearly ($B_{\text{tissue}} \propto M$), but the vascular tree limits delivery. West, Brown & Enquist [11] showed $Q_0 \propto M^{3/4}$ when area-preserving constraints at the capillary level are imposed. From the MRP perspective:

1. Murray's Law (from MRP) determines the radius ratio: $r_{\ell+1}/r_\ell = n^{-1/3}$.
2. Space-filling determines the length ratio: $l_{\ell+1}/l_\ell = n^{-1/d}$.
3. Capillary invariance fixes the terminal scale.
4. Blood volume $\propto M$ (empirical constraint).

Together: $N_T \propto M^{d/(d+1)}$ and therefore

$$\boxed{B \propto M^{d/(d+1)}.} \tag{18}$$

For $d = 3$: $B \propto M^{3/4}$—Kleiber's Law [12], now a consequence of MRP.

The exponent decomposes as $d/(d+1) = (d-1)/d + 1/[d(d+1)]$. The extra $1/[d(d+1)] = 1/12$ arises because the terminal impedance Z_L is fixed (capillaries are size-invariant), adding an additional boundary condition ($2/3 + 1/12 = 3/4$).

a. Volumetric heat density and the K-bottleneck. By Corollary 1 of Paper 0 [1], every active interface dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat. A tissue with K modes per node packs K such interfaces into a single cross-section; hence the *volumetric heat density* scales as

$$q \;\propto\; K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}/V_{\text{node}}\,. \tag{19}$$

The human brain (K=5) comprises ∼2% of body mass but hosts the highest K tissue in the organism. Because $q \propto K$, the brain generates a disproportionate share of metabolic heat—the classical 20 W out of 80 W basal (∼25%)—not despite being small, but *because* its per-node mode count is largest. Kleiber's $M^{3/4}$ law sets the whole-body envelope; Eq. (19) sets the *intra-body* distribution of that heat, concentrating it where K is highest. This creates a physical necessity for a dedicated heat-removal architecture (the cerebral vasculature), linking directly to the dual-network model of Paper 2 [2].

VIII. DUAL-NETWORK PACKING AND THE OVERHEAD Ω

A. Two trees in one volume

A multicellular organism requires two space-filling networks [2]: the **vascular tree** ($Z_v = \Delta P/Q$, transport) and the **neural tree** ($Z_n = R_m + 1/j\omega C_m$, control). Both must reach every cell. If independent, $A^{\text{total}}_{\text{cut}} = A^v_{\text{cut}} + A^n_{\text{cut}}$. But they share a finite volume.

B. The packing theorem

Theorem 2 (Dual-Network Overhead)**.** *When two space-filling relay trees share a volume in d dimensions,*

$$A^{dual}_{cut} = A^v_{cut} + A^n_{cut} + A_{couple}, \quad A_{couple} > 0. \tag{20}$$

Proof. At co-terminal points, the two networks negotiate resource allocation. The coupling mismatch $\Gamma_{\text{couple}} = (Z^{\text{demand}}_n - Z^{\text{supply}}_v)/(Z^{\text{demand}}_n + Z^{\text{supply}}_v)$ is generically nonzero because each tree optimises independently. Hence $A_{\text{couple}} = N_T\langle\Gamma^2_{\text{couple}}\rangle > 0$. □

C. The metabolic overhead factor

$$\Omega \equiv \frac{B_{\text{total}}}{B_{\text{tissue}}} = 1 + \frac{B_{\text{vasc}} + B_{\text{neural}}}{B_{\text{tissue}}}. \tag{21}$$

For humans: $B_{\text{neural}}/B_{\text{total}} \approx 0.20$ (brain ≈ 20% of BMR for ∼ 2% of mass [14]); $B_{\text{vasc}}/B_{\text{total}} \approx 0.05$ (cardiac work). Therefore

$$\Omega \approx \frac{1}{0.75} \approx 1.33. \tag{22}$$

The neural network alone gives $\Omega_n = 1/(1 - 0.20) = 1.25$—remarkably close to $\beta = 2^{1/3} \approx 1.26$. We conjecture that this near-equality is not coincidental: the metabolic tax of adding a second network is of order $\beta - 1$, reflecting the cost of one additional level of hierarchical branching.

D. Why blood before nerve

C2 ($\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$) requires a carrier for x_{in}. For neural C2: x_{in} is carried by neurotransmitters, whose synthesis requires glucose and oxygen delivered by blood. Without the vascular network, $x_{\text{in}} = 0$ everywhere, $\Delta Z = 0$: the neural network cannot remodel. Embryology confirms: cardiac tube forms at day 21; neural tube closes at day 25–28. In the product $1_v \times 1_n$, the vascular "1" is the left factor.

IX. GEOMETRIC UNIVERSALITY OF β

The ratio $\beta = n^{1/d}$ is a geometric inevitability for any space-filling hierarchy—independent of molecular substrate.

A. Plants: single network, same β

Vascular plants lack a neural network but obey Murray's Law with $\beta = 2^{1/3}$ for binary branching [15], and their metabolic scaling follows $B \propto M^{3/4}$ [16]. This confirms β is a property of geometry, not biology.

B. Single-network organisms: $\Omega \approx 1$

Plants, lacking a neural network, have $\Omega_{\text{plant}} \approx 1$. **Prediction:** $\Omega_{\text{plant}} < \Omega_{\text{animal}}$ by a factor approaching $\beta \approx 1.26$—testable by cross-kingdom metabolic comparisons.

X. HIERARCHICAL ARCHITECTURE: THE TREE OF NETWORKS

Paper 5 [5] identifies twenty-nine Γ-networks in the human body. They form a *tree*, not a flat list:

Level	Networks
Root	MRP (variational principle)
Controllers	Cardiac pump, Regulatory brain
Trunk	Vascular tree, Neural tree
Leaf	17 tissue networks
	(bone, muscle, immune, GI, . . .)

Every leaf network requires both trunks: x_{in} delivery from the vascular tree, η regulation from the neural tree.

The total action:

$$\begin{aligned} A_{\text{life}} &= \underbrace{A_v + A_n}_{\text{trunk}} + \underbrace{\textstyle\sum_{t=1}^{17} A_t}_{\text{leaves}} + \underbrace{A_{\text{couple}}}_{\text{packing}} \\ &= A^{\text{total}}_{\text{imp}}(t) + A^{\text{total}}_{\text{cut}}, \end{aligned} \tag{23}$$

where $A_{\text{imp}}^{\text{total}} \to 0$ under C2 across all networks, and

$$A_{\text{cut}}^{\text{total}} = A_{\text{cut}}^{v} + A_{\text{cut}}^{n} + \sum_{t=1}^{17} A_{\text{cut}}^{t} + A_{\text{couple}}. \tag{24}$$

This is the complete geometric structure of the irreducible cost of life.

XI. DISCUSSION

A. Methodological caveat

All numerical experiments in Part A use simulation modules built from C1–C3 themselves. When the simulator confirms a theorem derived from those axioms, it demonstrates *internal consistency*, not *empirical validity*. Only external measurements on biological impedance networks can promote these results to physical truth. The Part B scaling laws are mathematical consequences of the irreducibility theorem plus geometric constraints; no simulation is involved.

B. The irreducible cost as a design constraint

The brain must manage A_{cut} through *architectural* mechanisms rather than impedance remodeling. This explains: (1) *hierarchical processing*—cortical areas arranged in layers of decreasing K, with relay nuclei (e.g., thalamus, $K \approx 3$) mediating between cortex ($K = 5$) and periphery ($K = 1$); (2) *modular connectivity*—same-K connections strongly favoured within cortical columns; (3) *small-world topology*—diameter-2 structure enabling relay-mediated transmission.

C. C2 as a dimensional selector

Even without explicit soft cutoff ($\gamma = 0$), C2 impedance remodeling produces $D_K \approx 0.66$—an intrinsic dimensional preference. Same-K edges have more shared modes for gradient descent, creating higher survival probability under edge pruning. The soft-cutoff parameter γ then acts as a knob on top of this C2 bias, enabling continuous control of D_K from ~ 0.66 to ~ 2.3.

D. Functional regions as thermodynamic phase separation

The dimensional preference derived above has a direct anatomical corollary. Consider a homogeneous network of neurons with randomly assigned K values. C2 gradient descent on the reflection action $\mathcal{A}$ selectively strengthens same-K edges (more shared modes $\Rightarrow$ faster Γ reduction $\Rightarrow$ lower $\mathcal{A}$) and weakens cross-K edges (excess modes totally reflected, no gradient available). Under sustained activity plus edge pruning (developmental apoptosis), the surviving connectivity partitions the network into *same-K clusters*: neurons sharing a common modal dimension self-segregate into spatially contiguous aggregates—precisely the *functional regions* observed in cortical anatomy.

The argument is a thermodynamic phase-separation: MRP drives the system toward the lowest attainable $\mathcal{A}$; same-K clustering minimises A_{cut} at every internal edge. The result is a hierarchy of clusters:

- **Sensory cortices** ($K \approx 4$–8): neurons tuned to specific modalities (visual, auditory, somatosensory) cluster because their shared modal dimension enables maximal C2 convergence.
- **Motor cortex** ($K \approx 5$): output neurons with matched dimensionality aggregate into topographic motor maps.
- **Prefrontal cortex** ($K \approx 128$): the convergence point of all modalities; its extreme K forces it into a separate cluster because no lower-K region can absorb its modes without A_{cut} penalty.
- **Subcortical nuclei** ($K \approx 1$–3): relay stations (thalamus $K{\approx}3$, brainstem $K{\approx}2$) self-organise as intermediate clusters that bridge the K-gap between periphery and cortex.

Functional brain regions are therefore not a biological design choice but a *theorem consequence*: any impedance network under MRP with heterogeneous K *must* phase-separate into same-K domains to minimise total reflected action. Cortical columns—within which same-K connectivity is strongly favoured (point 2 above)—are the mesoscopic realisation of this thermodynamic phase.

E. Negative scaling exponent and brain-scale feasibility

$\tau_{\text{conv}} \sim N^{-1.12}$ predicts brain-scale networks ($N \sim 10^{10}$) converge in $\sim 10^{-11}$ ticks—effectively instantaneous. The mechanism is physically analogous to the Curie-Weiss model: effective field strength grows with system size.

F. What "$1 \times 1 = \Omega$" means

Each "1" is a topologically one-dimensional tree in 3D space. Embedding imposes fractal cost; packing two such trees adds $A_{\text{couple}} > 0$. The result $\Omega \approx 1.25$–1.33 is determined by the brain's metabolic share. The match $\Omega_n \approx \beta$ suggests: *the cost of adding a second hierarchical tree is of order one branching level of the first tree.*

G. Limitations

1. The self-similar mode profile is an idealisation; real networks have heterogeneous branching. The $M^{(d-1)/d}$ scaling is a leading-order result.

2. The identification $\Omega_n \approx \beta$ is empirically suggestive but not rigorously derived.

3. The capillary invariance constraint is an empirical input. Whether MRP alone determines the terminal scale is open.

4. The negative scaling exponent $\alpha = -1.12 \pm 0.12$ is measured at four sizes only ($N \leq 128$). Brain-scale extrapolation should be treated as a conjecture.

XII. FALSIFIABLE PREDICTIONS

1. **Cortical fractal dimension.** DTI-based grouping of axons by estimated mode count K should yield a connectivity decay $p(\Delta K) \propto (\Delta K + 1)^{-\gamma}$ with $\gamma \approx 0.7$–0.9, matching $D_K \in [1.3, 1.5]$ via Eq. (9).

2. **Vascular fractal dimension.** 3D micro-CT should yield box-counting dimensions $d_H \to 3$ below capillary spacing (space-filling) and lower d_H above (matching the effective tissue dimension).

3. **Ω across vertebrates.** Brain metabolic fraction should correlate with neural complexity: $\Omega_{\text{fish}} < \Omega_{\text{reptile}} < \Omega_{\text{mammal}} < \Omega_{\text{primate}}$.

4. **Murray ratio β invariance.** Vascular bifurcation ratio $\beta = 2^{1/3} \approx 1.26$ across all binary-branching species (mammals, birds, fish, plants), independent of body size.

5. **$\Omega \approx 1$ for plants.** Plants lack a neural network; their metabolic overhead should yield $\Omega_{\text{plant}} \approx 1.05$.

6. **Dual-network failure cascade.** Cardiac arrest (loss of vascular trunk) degrades all 17 tissue networks within minutes. Brain death (loss of neural trunk) degrades tissue networks on hours-to-days timescale.

7. **Embryonic fractal geometry.** Fetal vasculature should show Murray-optimal bifurcation ratios $r_p/r_d \approx \beta$ from the earliest stages of vasculogenesis (day 17–18), not only after postnatal maturation.

8. **Cartilage as relay node.** Articular cartilage impedance should satisfy $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} Z_{\text{syn}}}$, maximising transmission (Corollary 1). Osteoarthritic cartilage ($\eta \to 0$) should show increased Γ^2 and decreased T.

XIII. CONCLUSION

This paper establishes five results for heterogeneous impedance networks under the Minimum Reflection Principle.

First, the *Irreducibility Theorem* [Eq. (6)]: $A = A_{\text{imp}}(t) + A_{\text{cut}}$, where $A_{\text{imp}} \to 0$ under C2 impedance remodeling but A_{cut} is invariant under all local impedance rules at fixed topology.

Second, *negative scaling* [Eq. (7)]: $\tau_{\text{conv}} \sim N^{-1.12}$—convergence accelerates with network size.

Third, *fractal cost* [Eq. (13)]: $A_{\text{cut}} \propto M^{(d-1)/d} = M^{2/3}$ for $d = 3$, quantifying how big the irreducible cost is.

Fourth, the Murray bifurcation ratio $\beta = 2^{1/3} \approx 1.26$ and the dual-network overhead $\Omega \approx 1.25$–1.33 are geometric consequences of space-filling plus MRP.

Fifth, Kleiber's allometric law $B \propto M^{3/4}$ is recovered when MRP-derived branching is combined with space-filling and capillary invariance, providing a variational foundation for the WBE model.

The title equation is now precise:

$$1_v \times 1_n = \Omega \approx \beta = 2^{1/3} \approx 1.26. \tag{25}$$

Life pays a fractal tax of $\beta - 1 \approx 26\%$—the price of one extra branching level, expressed as the brain's share of the metabolic budget—for the privilege of being simultaneously fed and informed.

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